

Optimization of a UAV for Wildfire Management

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(ABSTRACT)

Give a brief description of your thesis here.

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(GENERAL AUDIENCE ABSTRACT)

You are also required as of Spring 2016 to include a general audience abstract. This should be geared towards individuals outside of your field that may be reading seeking information about your work. You should avoid language that is particular to your field and clearly define any terms that may have special meaning in your discipline.

Contents

List of Figures	vi
List of Tables	vii
1 Introduction	1
1.1 One Section	1
1.1.1 A sub-section	2
1.2 Another Section	3
2 Review of Literature	5
3 UAV Design	6
3.1 UAV Geometry	6
3.2 Mission Profile	7
3.3 UAV Batteries	7
4 Finite Element Analysis	8
4.1 UAV Segment Model	8
4.2 Beam Element Formulation	9
4.3 Solvers	9

4.3.1	Static	9
4.3.2	Dynamic	9
4.3.3	Transient	10
4.4	Optimization	10
4.4.1	Objective Function and Design Variables	13
4.4.2	Constraints	13
5	Results	15
6	Conclusions	16
7	Summary	17
	Appendices	18
	Appendix A First Appendix	19
A.1	Section one	19
A.2	Section two	21
	Appendix B Second Appendix	23

List of Figures

3.1	Average outlet velocity magnitude given an average input velocity magnitude of 0.88 m/s.	6
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List of Tables

5.1		15
-----	-------	----

NLP is a field of computer science, artificial intelligence, and linguistics concerned with the interactions between computers and human (natural) languages.

σ is the eighteenth letter of the Greek alphabet, and carries the 's' sound. In the system of Greek numerals, it has a value of 200.

Chapter 1

Introduction

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1.1 One Section

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1.1.1 A sub-section

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Chapter 2

Review of Literature

Chapter 3

UAV Design

3.1 UAV Geometry

- The UAV is modeled as a simplified version of a UAV frame

Images/part stuio 1.png

Figure 3.1: Average outlet velocity magnitude given an average input velocity magnitude of 0.88 m/s.

- Talk about the breifly original UAV design (I do not know where to find this)
- Frame designed from 3 components - Arms - Annulus - Design variables are radius, thickness
- Arm length runs from the end of the hub to the wingspan - the wingspan is the minimum required to fit all propellers with 25 mm of spacing - Struts - Annulus - Design variables are radius, thickness, and distance from center - Hub - Two octagonal plates with sides perpendicular to the arm - the web runs in between each
- Flange thickness, web thickness, and Hub radius are design variables.
- They are assumed to be welded together
- Payload of the UAV is 45 kg
- Material Properties - Table of the relevant material properties of dragon plate.
- Cross Section Properties - Table of the relevant cross section properties

- Arm length is assumed to be the minimum allowable length that allows for 50mm of spacing between propellers.
- Hub height is determined to be 64mm to allow for clearance of the battery. [battery source]

3.2 Mission Profile

= The goal of the mission is to go down - Mission profile is broken into segments - climb - methodology used to create the mission profile - The objective was to create a mission profile to describe a potential UAV mission with continuous position, velocity and acceleration - Constraints are set up on the position velocity and acceleration at the nodes between mission profile segments - a matrix of kinematic equations are used to determine the position and acceleration at the nodes - [reference book] - between mission segments nodes the position, velocity, and acceleration are interpolated using a cubic hermite spline - [reference]

3.3 UAV Batteries

<https://genstattu.com/tattu-40000mah-6s-10c-22-8v-high-voltage-uav-lipo-battery-pack-with-as150-as150-plug/?srsltid=AfmBOoo89wjweOS5x2-3PrUxTjnsa3qDuCLy9eYPu2DzErwfbAG77> battery dimensions : $4.292e - 3$,

"battery_dimensions" : $np.array([260.5, 123.5, 63.5])$,

"battery_volts" : 22.8,

"battery_mah" : 40000

Chapter 4

Finite Element Analysis

In order for the optimization algorithm to run repeatedly, a finite element beam model was created. The finite element solver is based on a Timoshenko beam theory. This formulation accounts for transverse shear deformation, which is important when analyzing beams with a short cross-section.

4.1 UAV Segment Model

- The FEA model is a single slice of the UAV with symmetric constraints at the midpoints of the strut beams. - Each beam is defined as an annular cross section. - The hub is represented as a series of repeated I-beam cross sections. - The width of the cross section is calculate to be the centroid of the segment

Each hub beam is labeled "hub_{beam_x}." *The number of hub sections is arbitrary and is increased until the solution of the beam. Finally, an annular arm beam spans from the point where the strut and arm meet to the outer*

[include figure]

4.2 Beam Element Formulation

- Boundary Conditions - Boundary conditions are set relative to a local coordinate system so that the symmetric boundary conditions at the endpoints of the struts can be set
- Stiffness Matrix Assembly

Each beam element is a Timoshenko beam. This beam model takes shear deformation effects into account. The mass for each beam is modeled as a 3D lumped mass matrix.

4.3 Solvers

The FEA model has three separate solvers. The static solver, which is used to calculate the response given a static loading, the dynamic solver, which calculate the response given a force of an oscillating force at a given frequency, and the transient solver which calculates the response given a time-changing force.

4.3.1 Static

The static solver takes an input of a point loads at given points. The static solver outputs the bending stresses and strains experienced by the beams.

*

4.3.2 Dynamic

[source]

4.3.3 Transient

[source]

4.4 Optimization

The problem is formulated as follows:

$$\min \text{mass}(d_a, t_a, d_s, t_s, r_s, t_f, t_w)$$

subject to:

$$g_j(x) \leq 0; j = 0, 1, 2, \dots, J$$

$$x_i^L \leq x_i \leq x_i^U$$

design variables:

- d_a : arm diameter,
- t_a : arm thickness,
- d_s : strut diameter,
- t_s : strut thickness,
- r_s : strut distance from center,
- t_f : hub flange thickness,

- t_w : hub web thickness
- h_l : hub length

derived variables:

- d_{tip} : Deflection at tip
- σ_{max} : Maximum stress from bending
- f_{amp} : Amplitude of displacement at the tip of the uav arm
- D : Damage per mission
- E_{used} : Energy used over th duration of the mission
- E_{total} : Energy used over th duration of the mission

design parameters:

- $d_{allowable}$: Deflection allowable
- $\sigma_{allowable}$: Stress allowable from bending
- $R_{allowable}$: Allowable amplitude displacement ratio
- $n_{missions}$: numbr of missions before damage

Constraints:

Limit on deflection from bending by force required to lift the UAV.

$$g_1(d_a, t_a, d_s, t_s, r_s, t_f, t_w, h_l) = d_{tip} \leq d_{allowable}$$

Limit on stress from bending by force required to lift the UAV.

$$g_2(d_a, t_a, d_s, t_s, r_s, t_f, t_w, h_l) = \sigma_{max} \leq \sigma_{allowable}$$

Limit on amplitude of displacement given a forcing function based on the propeller speed of the UAV.

$$g_3(d_a, t_a, d_s, t_s, r_s, t_f, t_w, h_l) = \frac{f_{amp}}{d_{tip}} \leq R_{allowable}$$

Limit on the damage the UAV during a mission.

$$g_4(d_a, t_a, d_s, t_s, r_s, t_f, t_w, h_l) = D * n_{missions} \leq 1$$

Energy

$$g_5(d_a, t_a, d_s, t_s, r_s, t_f, t_w, h_l) = E_{used}/E_{total} \leq 0.8$$

The bounds on design variable were determined by limits on commercially available carbon fiber based on the dragon plate ebsite.

$$9\text{mm} \leq r_a \leq 50.0\text{mm}$$

$$0.254\text{mm} \leq t_a \leq 3.175\text{mm}$$

$$9\text{mm} \leq r_s \leq 50.0\text{mm}$$

$$0.254\text{mm} \leq t_s \leq 3.175\text{mm}$$

$$400\text{mm} \leq d_s \leq 1127.3\text{mm}$$

$$100\text{mm} \leq h_l \leq 1127.3\text{mm}$$

"*arm_{diameter}*" : (9, 50), "*arm_{tthickness}*" : (0.254, 3.175), "*strut_{diameter}*" : (9, 50), "*strut_{tthickness}*" : (0.254, 3.175), "*strut_{distance}*" : (400, 1127.3), "*hub_{radius}*" : (100, 400), "*hub_{flange_{tthickness}}*" : (0.7, 12.7), "*hub_{web_{tthickness}}*" : (0.7, 3.175)

- Optimization algorithm used is SLSQP method using scypi's minimization function.

4.4.1 Objective Function and Design Variables

4.4.2 Constraints

Displacement

$$g_1(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = d_{tip} \leq d_{allowable}$$

Stress

$$g_2(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = \sigma_{max} \leq \sigma_{allowable}$$

Frequency

$$g_3(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = \frac{f_{amp}}{d_{tip}} \leq R_{allowable}$$

Damage

This mission profile with a rev up and rev down time of 5 seconds was added to the mission profile.

$$- \text{"mission_count_damage"} : 10000, g_4(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = D * n_{missions} \leq 1$$

Energy

$$g_5(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = E_{used}/E_{total} \leq 0.8$$

Chapter 5

Results

design variables $arm_diameter : 50.0$ $arm_thickness : 0.30233957405287726$ $strut_diameter :$
 50.0 $strut_thickness : 0.3609053941590715$ $strut_distance : 556.4454900912388$ $hub_radius :$
 100.0 $hub_flange_thickness : 0.7$ $hub_web_thickness : 0.7$ $constraintsdeflection :$
 0.9999999999999993 $stress : 0.977859849282712$ $displacementamplitude : 0.290648859567$
 0.3260612162830091 $damage : 0.0$ $massw/payload : 0.8250115057269486kg$

Table 5.1

Chapter 6

Conclusions

Chapter 7

Summary

Appendices

Appendix A

First Appendix

A.1 Section one

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A.2 Section two

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Appendix B

Second Appendix

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